Author resolution. The homogeneous degree-one affine-line and point classes, the degree-zero unit, and the degree-one generators of the graded ideal are now explicit. This records the existing grading convention and is applied under the author's standing authorization.

Let $!I$ be the ideal of $\mathbf{K}\,\mathrm{RES}$ generated by the elements $[\gamma^\sharp] - [\mathbb{G}_m]$, where $\gamma \in \Gamma$ is definable. The quotient ring $\mathbf{K}\,\mathrm{RES}/!I$ is written as $!\mathbf{K}\,\mathrm{RES}$. In the graded ring, $!I[*]$ is generated by the homogeneous degree-one differences $[\gamma^\sharp] - [\mathbb{G}_m] \in \mathbf{K}\,\mathrm{RES}[1]$, and we set $!\mathbf{K}\,\mathrm{RES}[*] = \mathbf{K}\,\mathrm{RES}[*]/!I[*]$.

The ideal of $\mathbf{K}\,\mu\mathrm{RES}[*]$ generated by the elements

$$[(\mathbb{G}_m, \alpha + \beta)] - [(\alpha^\sharp, \beta)] \in \mathbf{K}\,\mu\mathrm{RES}[1],$$

where $\alpha, \beta \in \Gamma$ are definable, is denoted by $!\mu I[*]$, and the (graded) quotient ring $\mathbf{K}\,\mu\mathrm{RES}[*]/!\mu I[*]$ by $!\mathbf{K}\,\mu\mathrm{RES}[*]$.

The quotient maps from "$\mathbf{K}$" to "$!\mathbf{K}$" will all be denoted by ι, and will often be omitted when it is clear in the context that the latter one is meant. We will also use the same notation for elements when passing from the former to the latter.

Note that the category $\mu\mathrm{RES}[*]$ and the ring $!\mathbf{K}\,\mu\mathrm{RES}[*]$ will be dismissed after Proposition 4.7 as inadequate for our purpose, and a new construction is introduced thereafter.

4.1. **Without volume forms.** The discussion here more or less follows [9, § 2.5] (also see [8, Theorem 10.5]).

For $0 \leq l \leq n$, consider the maps

$$\epsilon_g^{n,l} : \mathbf{K}\,\mathrm{RES}[n-l] \otimes_{\mathbb{Z}} \mathbf{K}\,\Gamma[l] \longrightarrow !\mathbf{K}\,\mathrm{RES}[n]$$

given by

$$x \otimes y \longmapsto \chi_g(y) x [\mathbb{G}_m]^l. \tag{4.1}$$

The component maps are first defined on tensor products over $\mathbb{Z}$. Their sum respects the homogeneous balancing relations over $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$ after passing to $!\mathbf{K}\,\mathrm{RES}[*]$: a finite definable Gamma set is a disjoint union of its definable singleton values, and moving such a singleton between the two factors changes its contribution from $[\gamma^\sharp]$ to $[\mathbb{G}_m]$, whose difference vanishes in the target. The resulting tensor assignment also identifies the coefficient-translation pairs of Proposition 3.18. Indeed, if a RES coefficient R admits a definable map $R \longrightarrow \delta^\sharp$, the corresponding pair translates a single coordinate of the Γ factor by δ. Both χ_g and χ_b are invariant under that definable translation. Thus the degree-n sum descends through the kernel congruence of the surjection Ψ, giving a homomorphism

Resolved referee R058 [Clarification] – Tensor products of homogeneous pieces.
Original source line 1442. Strictly speaking, an individual homogeneous piece such as $\mathbf{K}\,\mathrm{RES}[n-l]$ is not a module over the whole graded ring $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$. Define the displayed component maps first on $\mathbf{K}\,\mathrm{RES}[n-l] \otimes_{\mathbb{Z}} \mathbf{K}\,\Gamma[l]$, then show that their sum descends through the homogeneous balancing relations in the full graded tensor product. The quotient $!I[*]$ gives exactly the required compatibility.

Author resolution. The component maps are now explicitly over the integers; the subsequent sum is shown to respect the full graded balancing relations. This is a clarification of the construction, applied under the standing instruction for clear corrections.

$$\mathbf{K}\,\mathrm{RV}[n] \longrightarrow !\mathbf{K}\,\mathrm{RES}[n].$$

For later use, denote the resulting graded homomorphisms, before degree normalization, by

$$\widetilde{\mathbb{E}}_g, \widetilde{\mathbb{E}}_b : \mathbf{K}\,\mathrm{RV}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*], \qquad \widetilde{\mathbb{E}}_\chi\big(\Psi(x \otimes y)\big) = \chi(y) x [\mathbb{G}_m]^l \quad (y \in \mathbf{K}\,\Gamma[l]).$$

No further quotient of the RES target is required for this descent. Composing this with the homomorphism

$$\epsilon_g^{n,m} : !\mathbf{K}\,\mathrm{RES}[n] \longrightarrow !\mathbf{K}\,\mathrm{RES}[m] : x \longmapsto x[\mathbb{A}]^{m-n}$$

and taking the direct sum of the resulting homomorphisms over all n with $n \leq m$, we obtain a homomorphism

$$\epsilon_g^{m} : \mathbf{K}\,\mathrm{RV}[\leq m] \longrightarrow !\mathbf{K}\,\mathrm{RES}[m].$$

These homomorphisms ϵ_g^m, $m = 0, 1, \ldots$, are compatible in the obvious sense. Consequently, they give rise to a homomorphism

$$\mathbb{E}_g : \mathbf{K}\,\mathrm{RV}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}]$$

whose range is precisely the zeroth piece $(!\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}])_0$ of the $\mathbb{Z}$-graded ring on the righthand side. We have

$$\mathbb{E}_g([\mathrm{RV}^{\circ\circ}] + 1 - [1]) = -[\mathbb{G}_m][\mathbb{A}]^{-1} + 1 - [1][\mathbb{A}]^{-1} = 0. \tag{4.2}$$

All terms have degree zero: the degree-one point maps to $[1][\mathbb{A}]^{-1}$. The cancellation uses the homogeneous identity $[\mathbb{A}] = [\mathbb{G}_m] + [1]$.

Resolved referee R059 [Error] – Missing degree-one point factor.
Original source line 1461. In the graded target, $\mathbb{E}_g([1]) = [1][\mathbb{A}]^{-1}$, not $[\mathbb{A}]^{-1}$. Thus the correct computation is $-[\mathbb{G}_m][\mathbb{A}]^{-1} + 1 - [1][\mathbb{A}]^{-1} = 0$, using the homogeneous identity $[\mathbb{A}] = [\mathbb{G}_m] + [1]$. As written, the final term has degree -1, while the first two have degree 0. The formula $\mathbb{E}_g([1]) = [\mathbb{A}]^{-1}$ becomes correct only after forgetting the grading.

Author resolution. The missing degree-one point factor is restored. The corrected cancellation follows from the homogeneous affine-line identity; its algebraic calculation is checked in Lean.

Thus $\mathbb{E}_g$ induces an eponymous homomorphism

$$\mathbb{E}_g : \mathbf{K}\,\mathrm{RV}[*]/(\boldsymbol{P} - 1) \longrightarrow !\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}].$$

There is a parallel construction, where we replace χ_g with χ_b and $[\mathbb{A}]^{-1}$ with $[1]^{-1}$. The latter replacement is needed so to make $(\boldsymbol{P} - 1)$ vanish as in (4.2). The resulting homomorphism is

$$\mathbb{E}_b : \mathbf{K}\,\mathrm{RV}[*]/(\boldsymbol{P} - 1) \longrightarrow !\mathbf{K}\,\mathrm{RES}[*][[1]^{-1}], \tag{4.3}$$

whose range is precisely the zeroth piece $(!\mathbf{K}\,\mathrm{RES}[*][[1]^{-1}])_0$.

Remark 4.4. The zeroth graded piece $(!\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}])_0$ of $!\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}]$ is canonically isomorphic to the colimit of the groups $!\mathbf{K}\,\mathrm{RES}[n]$ with the transition morphisms $\epsilon_g^{n,m}$, which is actually what appears in the construction above. So there is a natural homomorphism from $(!\mathbf{K}\,\mathrm{RES}[*][[\mathbb{A}]^{-1}])_0$ to the similar-looking ring $!\mathbf{K}\,\mathrm{RES}[[\mathbb{A}]^{-1}]$, induced by the assignment $x \longmapsto x[\mathbb{A}]^{-n}$ for $x \in !\mathbf{K}\,\mathrm{RES}[n]$. Note that this homomorphism is not surjective.

Resolved referee R060 [Error; revised] – The asserted injection is false.
Original source line 1474. The natural homomorphism in this sentence need not be injective, already over an untwisted residue field $\mathbb{C}$. To distinguish the two constructions precisely, let R_n be the Grothendieck group of complex constructible sets of dimension at most n, with scissor relations imposed *within that category*, and set $R = \bigoplus_{n\geq 0} R_n$. This is the graded RES ring in the untwisted case; R_n is not by definition the nth dimension-filtration subgroup of the ordinary Grothendieck ring. Let $\mathbb{L}_1 = [\mathbb{A}^1]_1$. Ito–Miura–Okawa–Ueda give smooth projective threefolds X, Y that are not stably birational but satisfy $([X] - [Y])\mathbb{L} = 0$ in the ordinary ring (Theorems 1.1 and 1.3). Thus $z = ([X]_3 - [Y]_3)\mathbb{L}_1^{-3}$ maps to zero in $K_0(\mathrm{Var}_{\mathbb{C}})[\mathbb{L}^{-1}]$. Yet $z \neq 0$: for every $m \geq 0$, the additive invariant on R_{3+m} recording its top-dimensional birational classes distinguishes $X \times \mathbb{A}^m$ and $Y \times \mathbb{A}^m$. Hence no homogeneous power of $\mathbb{L}_1$ kills the numerator. To fit the paper's parameter conventions, take the full algebraically closed Puiseux

Resolved referee R063 [Typo] – Restore the degrees in the multiplicativity check.
Original source line 1544. The symbols q_α alone are not degree-zero elements of the displayed domain. State the check as $\psi(q_\alpha X^k)\psi(q_\beta X^l) = \psi(-q_{\alpha+\beta}X^{k+l})$ for $k, l \geq 1$. Also state explicitly that the second and third assignments in (4.5) apply only for $k \geq 1$; the separate interpretation at $k = 0$ applies to p_γ.

Author resolution. The positive-degree domain of the ray and constant assignments and the degrees in the multiplicativity check are now explicit.

Let $\mathfrak{h}$ denote the element $[(\mathrm{RV}^{\circ\circ}, -\mathrm{vrv})] \in \mathbf{K}\,\mu\mathrm{RV}[1]$.

Proposition 4.7. *There are two graded ring homomorphisms*

$$\mu\mathbb{E}_g, \mu\mathbb{E}_b : \mathbf{K}\,\mu\mathrm{RV}[*] \longrightarrow {!}\mathbf{K}\,\mu\mathrm{RES}[*]/([\mathbb{A}])$$

such that

- $\boldsymbol{P} \in \mathbf{K}\,\mu\mathrm{RV}[1]$ *vanishes under both of them,*
- *for all* $x \in \mathbf{K}\,\mu\mathrm{RES}[*]$, $\mu\mathbb{E}_g(x) = \mu\mathbb{E}_b(x) = x/([\mathbb{A}])$,
- *they are distinguished by* $\mu\mathbb{E}_g(\mathfrak{h}) = +[1]/([\mathbb{A}])$ *and* $\mu\mathbb{E}_b(\mathfrak{h}) = 0$.

Resolved referee R064 [Error] – Sign of the geometric retraction on h.
Original source line 1560. The construction gives the opposite sign in the third bullet. For $\boldsymbol{I} = ((0,\infty), -\mathrm{id})$, the corrected weight μ_I is identically zero, so $\lambda_g([\boldsymbol{I}]) = -p_0X$. Consequently $\mu\mathbb{E}_g(\mathfrak{h}) = \psi(-p_0X) = -[\mathbb{G}_m] = +[1]$ modulo $([\mathbb{A}])$. The bounded value is indeed zero. Replace $-[1]/([\mathbb{A}])$ by $+[1]/([\mathbb{A}])$. This confirms the sign concern in the supplied human referee report; repeating the sign printed in the cited HK theorem does not fix its incompatibility with the present definitions.

Author resolution. The sign is corrected directly from the stated construction: the corrected weight is zero, the geometric ray contributes $-p_0X$, and its image is $-[\mathbb{G}_m] = +[1]$ modulo the affine-line class. The bounded value remains zero.

Proof. The product map

$$(\iota/([\mathbb{A}])) \times (\psi \circ \lambda) : \mathbf{K}\,\mu\mathrm{RES}[*] \times \mathbf{K}\,\mu\Gamma[*] \longrightarrow {!}\mathbf{K}\,\mu\mathrm{RES}[*]/([\mathbb{A}])$$

is $\mathbf{K}\,\mu\Gamma^{\mathrm{fin}}[*]$-bilinear (because of the ideal ${!}\mu\boldsymbol{I}[*]$ in the target). It also identifies the measured coefficient-translation pairs of Proposition 3.18. For a translation $\eta = \gamma + \delta e_i$ of (I, σ), the target volume form is

$$\sigma_\delta(\eta) = \sigma(\eta - \delta e_i) - \delta.$$

Consequently

$$\sigma_\delta(\eta) + \Sigma\eta = \sigma(\gamma) + \Sigma\gamma.$$

The corrected-weight fibers are therefore definably bijective, so their two Euler characteristics agree and λ has the same value on the pair. The balanced product map thus descends through the kernel congruence of the surjection $\mu\Psi$, giving the desired graded homomorphisms. To compute their values, use the explicit tensor preimages

$$\mu\Psi\big([1] \otimes 1 - 1 \otimes [(0,\infty)]\big) = \boldsymbol{P}, \qquad \mu\Psi\big(1 \otimes [((0,\infty), -\mathrm{id})]\big) = \mathfrak{h}.$$

These formulas use preimages under the product map. For the value on $\mathfrak{h}$, the corrected weight on $((0,\infty), -\mathrm{id})$ is zero, so

$$\lambda_g([((0,\infty), -\mathrm{id})]) = -p_0X, \qquad \mu\mathbb{E}_g(\mathfrak{h}) = \psi(-p_0X) = -[\mathbb{G}_m] = +[1] \pmod{([\mathbb{A}])}.$$

For χ_b the open ray has characteristic zero, giving $\mu\mathbb{E}_b(\mathfrak{h}) = 0$.

Resolved referee R065 [Typo] – Agreement.
Original source line 1570. Change the final verb "follow" to "follows," since the subject is "The existence."

Author resolution. The subject–verb agreement is corrected under the author's standing instruction for clear corrections.

□

The discussion above is more or less an expanded version of the proof of [8, Theorem 10.11].

Remark 4.8. Alternatively, for $\chi = \chi_g$, we may replace $rX^k \longmapsto 0$ in (4.5) with $rX^k \longmapsto [\mathbb{G}_m]^{k-1}[1]$, that is, sending rX^k and q_0X^k to the same element. Let J be the enlargement of $!\mu I[*]$ by the elements $[1] - [(\{1\}, \gamma)] \in \mathbf{K}\,\mu\mathrm{RES}[1]$. Then ψ is still a graded ring homomorphism with this enlarged quotient. Consequently, for instance, we can construct a ring homomorphism

$$\mathbf{K}\,\mu\mathrm{RV}[*] \longrightarrow \mathbf{K}\,\mu\mathrm{RES}[*]/(J + ([\mathbb{A}])) \xrightarrow{\phi} !\mathbf{K}\,\mathrm{RES}[*]/([\mathbb{A}]), \tag{4.6}$$

where the homomorphism ϕ is induced by the obvious forgetful functor.

Resolved referee R066 [Clarification; revised] – Name the enlarged quotient.
Original source line 1580. Optional notation clarification: the preceding sentence already stipulates enlargement of $!\mu I[*]$. In the alternative construction, name the enlarged ideal J and write the middle term of (4.6) as $\mathbf{K}\,\mu\mathrm{RES}[*]/(J + ([\mathbb{A}]))$, so that the old and enlarged quotients are visibly distinguished. Read with that stipulated enlargement, the construction is consistent. The original assignment $rX^k \mapsto 0$ needs no such additional relations.

Author resolution. The already stipulated enlarged ideal is named J and the alternative target displays J explicitly; no additional relation is imposed on the original construction.

We shall not, however, pursue this further, since taking the quotient by the ideal $([\mathbb{A}])$ is detrimental to our purpose here. To avoid it, we need to work with the categories of doubly bounded objects.

Definition 4.9 (Doubly bounded RV-categories)**.** An object $(U, f) \in \mathrm{RV}[k]$ is *doubly bounded* if U is. Denote the full subcategory of $\mathrm{RV}[k]$ of doubly bounded objects by $\mathrm{RV}^{\mathrm{db}}[k]$, and similarly for the categories $\mu\mathrm{RV}^{\mathrm{db}}[k]$, $\mu\Gamma^{\mathrm{db}}[k]$.

Remark 4.10. By Lemma 2.41, if $(U, f, \omega) \in \mu\mathrm{RV}^{\mathrm{db}}[k]$ then $f(U)$, $\omega(U)$ are both doubly bounded as well. The product construction gives a homomorphism

$$\mu\Psi^{\mathrm{db}} : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes \mathbf{K}_+\,\mu\Gamma^{\mathrm{db}}[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}^{\mathrm{db}}[*].$$

We use it below to compute images of explicit tensor classes. The bounded retraction itself is obtained by restricting the already constructed graded map $\widetilde{\mathbb{E}}_b$ after forgetting volume forms.

Notation 4.11. Let $\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})$ be the subgroup of $\mathrm{FN}(\Gamma, \mathbb{Z})$ consisting of functions whose nonzero locus is doubly bounded. For each definable $\gamma \in \Gamma$, set

$$o_\gamma = \begin{cases} \mathbf{1}_{(0,\gamma)}, & \gamma > 0, \\ \mathbf{1}_{(\gamma,0)}, & \gamma < 0, \\ -p_0, & \gamma = 0. \end{cases}$$

As a $\mathbb{Z}$-module, $\mathrm{FN}^{\mathrm{db}}(\Gamma, \mathbb{Z})$ is generated by p_γ, o_γ. For $\gamma > 0$ we have $o_\gamma = q_0 - q_\gamma - p_\gamma$, and for $\gamma < 0$ we have $o_\gamma = q_\gamma - q_0 - p_0$. The convention $o_0 = -p_0$ is a convention for functions; it does not identify o_0 with the characteristic function of the empty interval $(0, 0)$. The convolution products are given by the following cases.

Resolved referee R067 [Error] – Negative interval generators and bounded support.
Original source line 1600. For $\gamma < 0$, the interval $(0, \gamma)$ is empty, so the definition as written cannot yield the claimed generators or quotient: the relation $o_\gamma + p_0$ would even kill the unit p_0. Define $o_\gamma = \mathbf{1}_{(\gamma,0)}$

Clearly $\boldsymbol{P}_\gamma$ does not depend on the choice of $t_\gamma \in \gamma^\sharp$. The ideal of $\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*]$ generated by the elements $\boldsymbol{P}_\gamma$ with $\gamma \in D_{\mathrm{RV}}$ is denoted by $(\boldsymbol{P}_\Gamma)$. A definable Gamma value need not have a definable RV lift, as already noted after Definition 3.13. All choices are made over the original parameter base. The images of $(\boldsymbol{P}_\Gamma)$ under the obvious forgetful homomorphisms

$$\mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow \mathbf{K}\,\mathrm{RV}[*], \quad \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow \mathbf{K}\,\mu\mathrm{RV}[*]$$

are contained in $(\boldsymbol{P}-1)$, $(\boldsymbol{P})$, respectively.

Resolved referee R070 [Typo] – Closing punctuation.
Original source line 1627. End this sentence with a full stop, not a comma.

Author resolution. The sentence now ends with a full stop.

Remark 4.14. Let $!P$ denote the ideal of $\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})$ generated by the elements $p_\gamma - p_0$, $o_\gamma + p_0$. By the computation in Notation 4.11, $\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})/!P \cong \mathbb{Z}$; the quotient homomorphism is denoted by χ because a simple computation shows that if $\boldsymbol{I} = (I,\mu) \in \mu\Gamma^{\mathrm{db}}[*]$ then indeed $\lambda_{\boldsymbol{I}}/!P = \chi(I)$. Now it is straightforward to check that the assignments

$$p_\gamma X^k \longmapsto [\mathbb{G}_m]^k, \quad o_\gamma X^k \longmapsto -[\mathbb{G}_m]^k \tag{4.7}$$

induce a graded ring homomorphism

$$\psi^{\mathrm{db}} : \mathbb{Z}[\Gamma_{\mathrm{def}}] \oplus X\,\mathrm{FN}^{\mathrm{db}}(\Gamma,\mathbb{Z})[X] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*].$$

In a nutshell, the homomorphism $\psi^{\mathrm{db}} \circ \lambda$ is a "forgetful" map given by $[(I,\mu)] \longmapsto \chi(I)[\mathbb{G}_m]^k$, $(I,\mu) \in \mu\Gamma^{\mathrm{db}}[k]$.

The forgetful homomorphism $\phi : \mathbf{K}\,\mu\mathrm{RES}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*]$ is similar to the one in (4.6).

Proposition 4.15. *There is a graded ring homomorphism*

$$\mu\mathbb{E}^{\mathrm{db}} : \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow !\mathbf{K}\,\mathrm{RES}[*]$$

such that $(\boldsymbol{P}_\Gamma)$ *vanishes and, for all* $x \in \mathbf{K}\,\mu\mathrm{RES}[*]$, $\mu\mathbb{E}^{\mathrm{db}}(x) = \phi(x)$.

Proof. Let $F : \mathbf{K}\,\mu\mathrm{RV}^{\mathrm{db}}[*] \longrightarrow \mathbf{K}\,\mathrm{RV}[*]$ be the forgetful homomorphism and set

$$\mu\mathbb{E}^{\mathrm{db}} = \widetilde{\mathbb{E}}_b \circ F.$$

This is graded, and its restriction to $\mathbf{K}\,\mu\mathrm{RES}[*]$ is ϕ. On the bounded tensor product its assignment is $\phi \times (\psi^{\mathrm{db}} \circ \lambda)$, by Remark 4.14. Thus it remains to compute the radius generators. If $\gamma = 0$, the annulus is empty and the value-zero singleton classes $[\{t_0\}]$ and $[1]$ agree, so $\boldsymbol{P}_0 = 0$. For an indexed radius $\gamma > 0$, we have

$$\mu\Psi^{\mathrm{db}}\big(1 \otimes [(0,\gamma)] + 1 \otimes [\{\gamma\}] + [\{t_\gamma\}] \otimes 1 - [1] \otimes 1\big) = \boldsymbol{P}_\gamma.$$

Its image under $\phi \otimes (\psi^{\mathrm{db}} \circ \lambda)$ is

$$\chi((0,\gamma))[\mathbb{G}_m] + \chi(\{\gamma\})[\mathbb{G}_m] + [1] - [1] = -[\mathbb{G}_m] + [\mathbb{G}_m] = 0.$$

The open-interval-plus-endpoint expansion is used only for $\gamma > 0$.

Resolved referee R071 [Error] – The zero-radius case in the proof.
Original source line 1657. The displayed decomposition and its vanishing computation apply only for $\gamma > 0$. At $\gamma = 0$, $\boldsymbol{P}_0 = 0$, but the displayed right side leaves the nonzero term $[\{0\}]$ in the $\Gamma[1]$ factor, whose image is $[\mathbb{G}_m]$. Handle $\gamma = 0$ separately, and for $\gamma > 0$ write the calculation explicitly: $\chi((0,\gamma))[\mathbb{G}_m] + \chi(\{\gamma\})[\mathbb{G}_m] + [1] - [1] = -[\mathbb{G}_m] + [\mathbb{G}_m] = 0$. The artificial convention $o_0 = -p_0$ concerns functions; it does not change the actual class of the empty interval $(0,0)$.